

Attention and transformers

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- 2 Visual attention
- 3 The transformer architecture
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Transformers: a new revolution in deep learning

- Transformers [Vaswani et al., 2017] have brought a break-through in natural language processing
- They have contributed to the development of new natural language processing applications (translation, voice assistants, etc.)
- They are now also extensively used in computer vision

What are transformers?

Definition

A transformer is a neural network architecture module that allows the network to **adaptively focus its attention** on certain regions of the data.

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Transformers today

Nowadays, when people refer to transformers, they generally mean the architectures based on the one introduced by Vaswani *et al.* in 2017 [Vaswani et al., 2017].

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One of the main ingredients in transformers is the attention mechanism.

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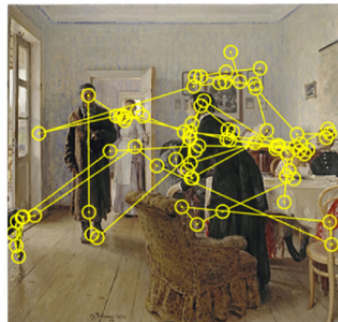
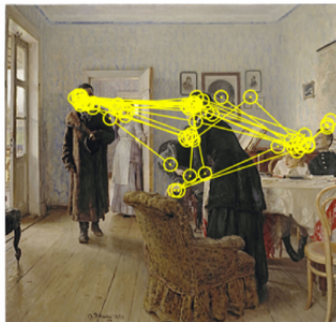
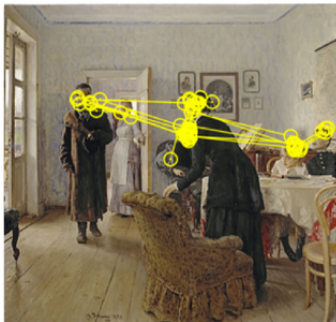
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How do we look at an image?



Credits: Ilya Repin, An Unexpected Visitor, 1884.

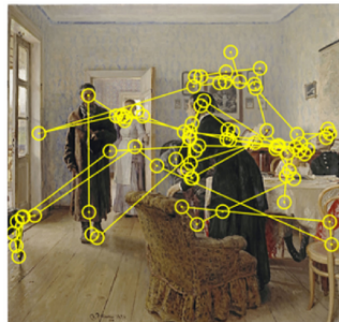
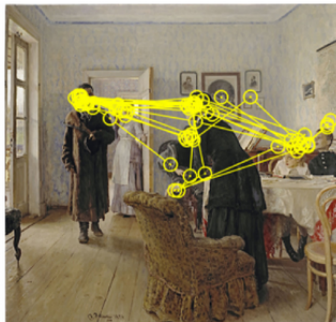
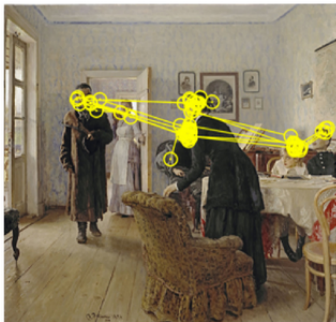
The attention path is linked to the task



Tasks:

- Age of the characters?

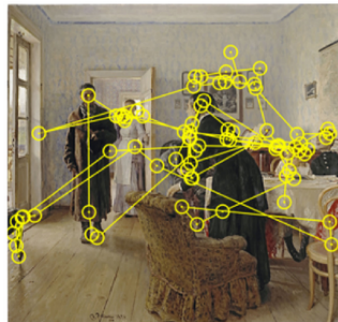
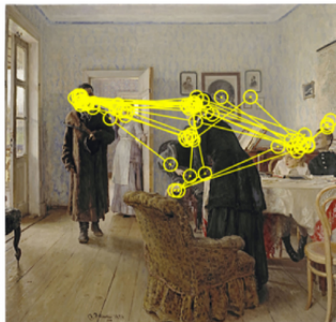
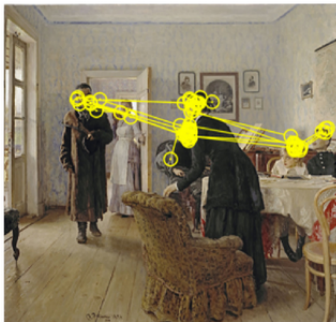
The attention path is linked to the task



Tasks:

- Age of the characters?
- How long has the visitor been away?

The attention path is linked to the task



Tasks:

- Age of the characters?
- How long has the visitor been away?
- Memorize the objects in the scene.

Human visual attention mechanisms

- Bottom-up:
 - local features (orientation, intensity, junctions, colour, motion, etc.)
 - local features contrast
 - context
- Top-bottom: task related
- Construction of a single *saliency map*

Exploring the image



- Winner-takes all! We focus on the maximum of the saliency map.
- Inhibition of return: We explore the following maxima, at first avoiding those that have already been inspected

Why has visual attention evolved?

- Photoreceptor cells are expensive
- Processing power is limited
- Solution: concentrate the cells in a given region and use the gaze to optimize their use

Why has visual attention evolved?

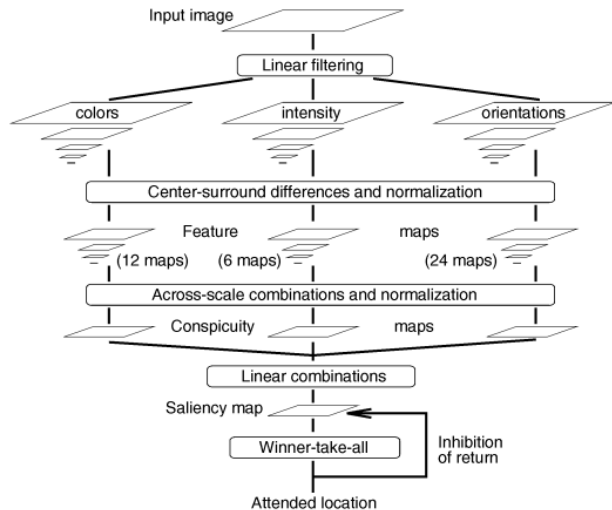
- Photoreceptor cells are expensive
 - Processing power is limited
 - Solution: concentrate the cells in a given region and use the gaze to optimize their use
- The same arguments apply to artificial visual systems

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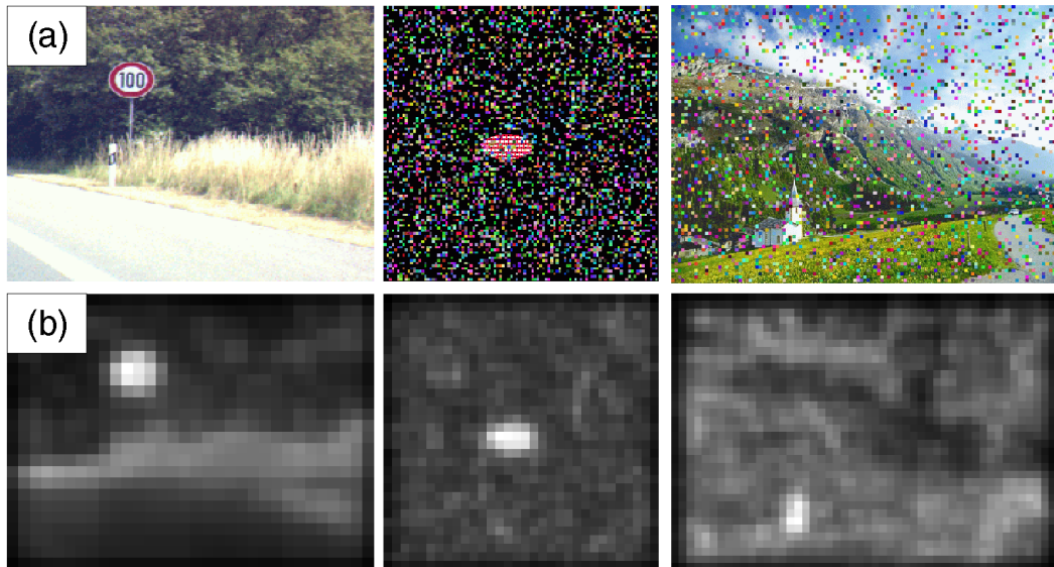
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A classical bottom-up model

- Itti et al. [Itti et al., 1998] proposed a model inspired by the primate visual system.
- It only uses low-level information.



Examples [Itti et al., 1998]

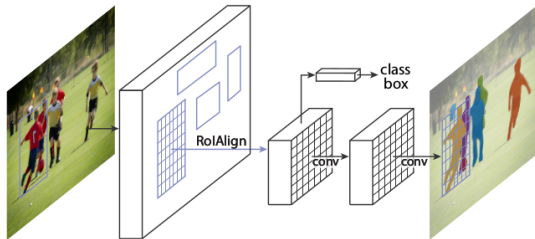


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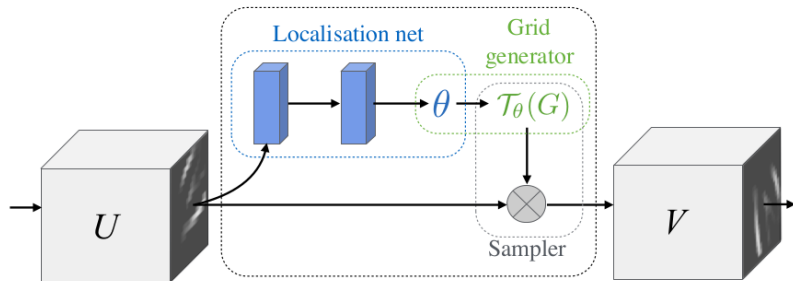
Region proposal networks [Ren et al., 2015]

- Detection and instance segmentation methods use region proposal networks, that can be interpreted as an attention mechanism.
- The region proposal network gives the coordinates of the rectangle and a probability that it contains an object.

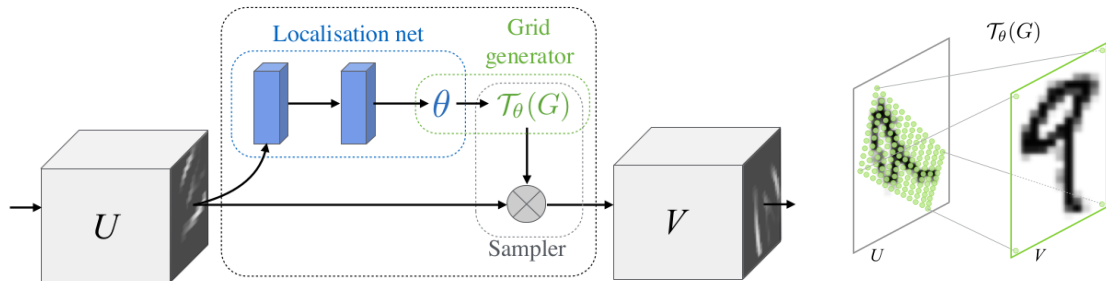


A region proposal module is used by mask R-CNN [He et al., 2017]

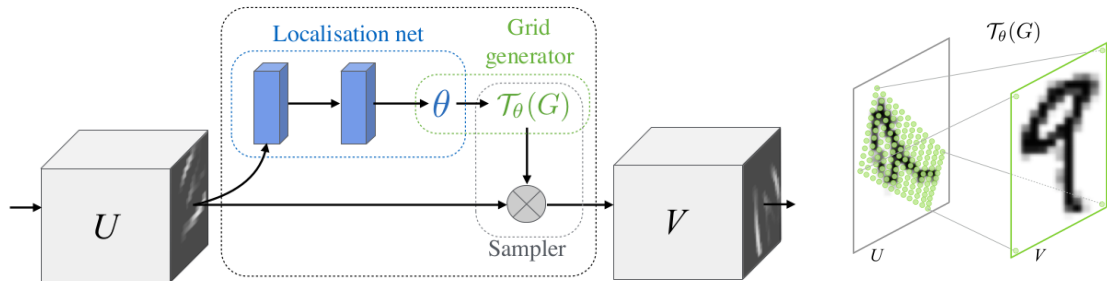
Spatial transformers [Jaderberg et al., 2016]



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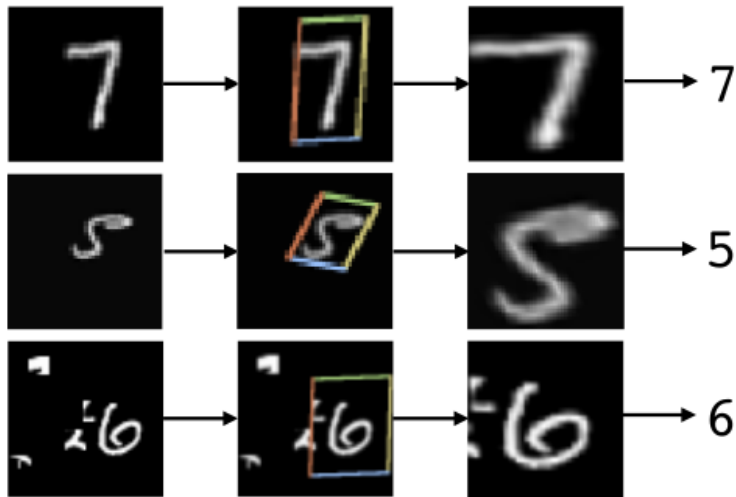


Spatial transformers [Jaderberg et al., 2016]

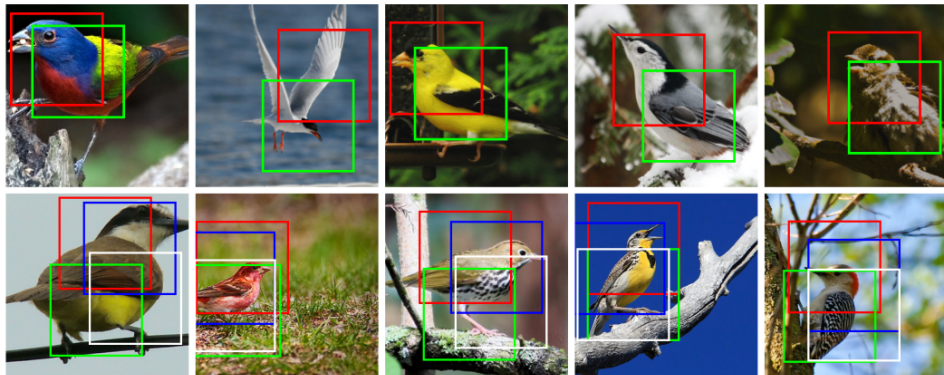


- This module can be added to any convolutional network
- End-to-end learning

Spatial transformers illustration



Spatial transformers with multiple heads



Remarks

- Note that in the first row one transformer tends to focus on the bird's head, while the second is centered on the body
- In the second row, the specialization is less apparent

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Transformer avatars

Some examples

- Graph transformers [Lecun et al., 1998]
- Transforming auto-encoders [Hinton et al., 2011]
- Spatial transformers [Jaderberg et al., 2016]

The transformer [Vaswani et al., 2017].

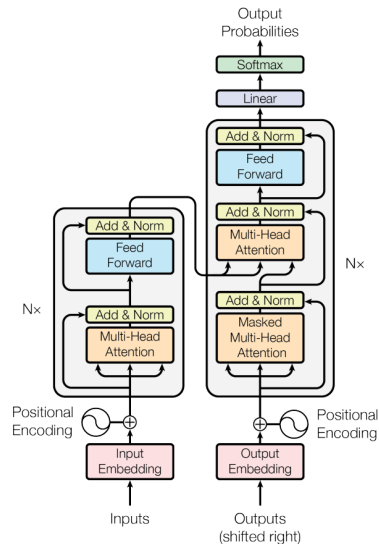
Today, when people refer to the transformer, they generally mean the architecture proposed by Vaswani et al. in 2017.

The rise of transformers

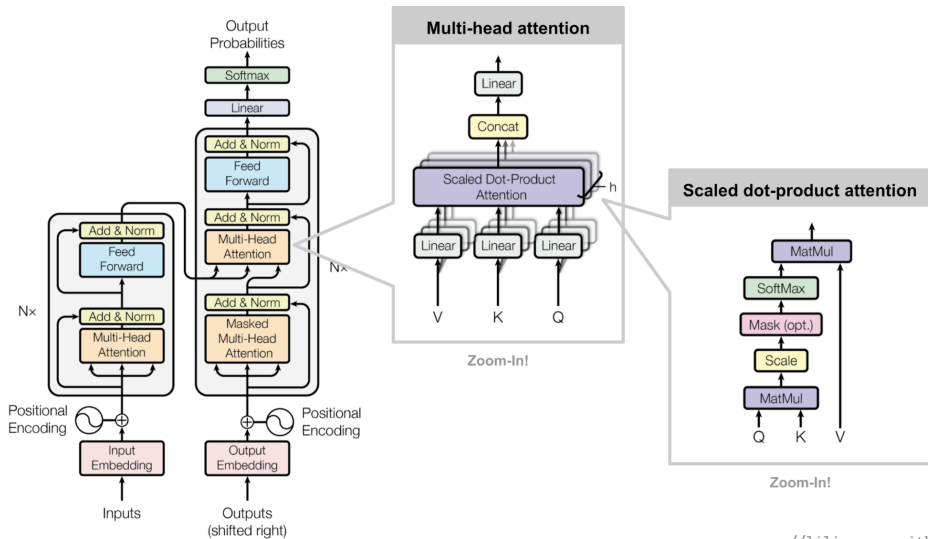
The paper that started it all

Vaswani et al., Attention is all you need, Neurips 2017.

This architecture was developed for text processing.



Architecture [Vaswani et al., 2017]



Credits: <https://lilianweng.github.io/lil-log/2018/06/24/attention-attention.html>

Success of transformers in natural language processing – some examples

- Bidirectional Encoder Representations from Transformers (BERT, by Google [Brown et al., 2020])
- Generative Pre-trained Transformer 3 (GPT-3, by OpenAI [Devlin et al., 2019]): 175 billion parameters ; GPT4 (2023) ; GPT5 (2025) ...
- Llama models, by Meta (Open Source)
- Le Chat, by Mistral AI
- etc...
- Chatbot arena: <https://lmarena.ai/>

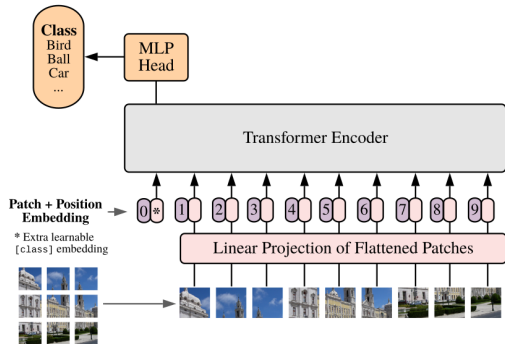
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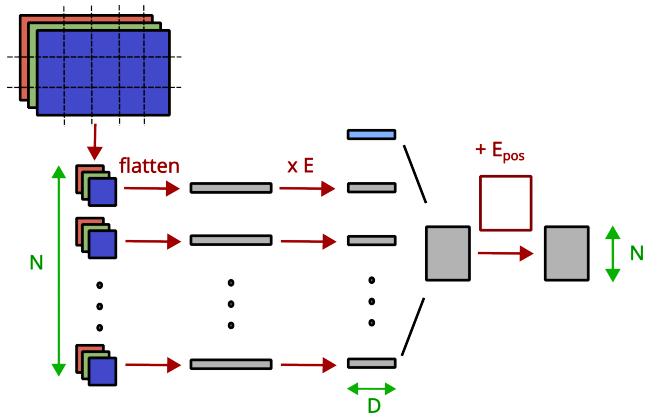
ViT: the vision transformer [Dosovitskiy et al., 2021]



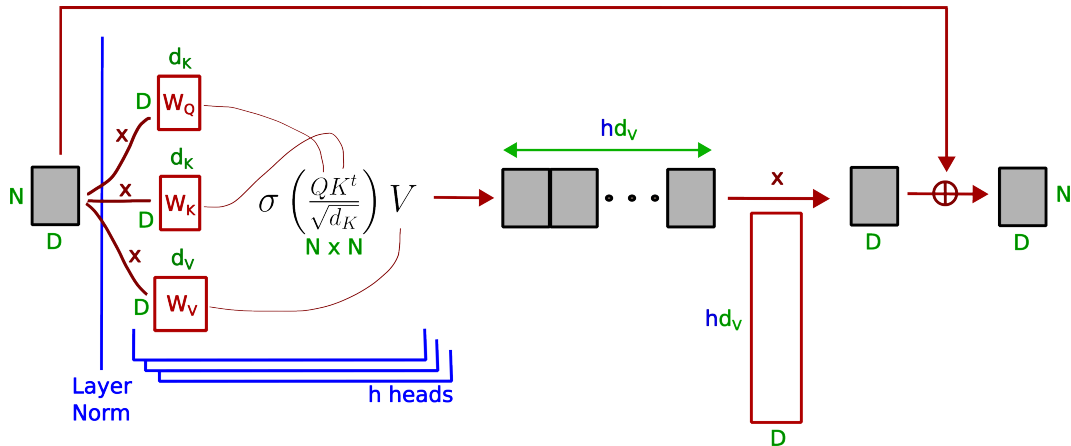
Remarks

- Only uses the transformer encoder
- Directly takes as inputs image patches
- Achieves state-of-the-art results when pre-trained on very large databases (Google's JFT-300M dataset)

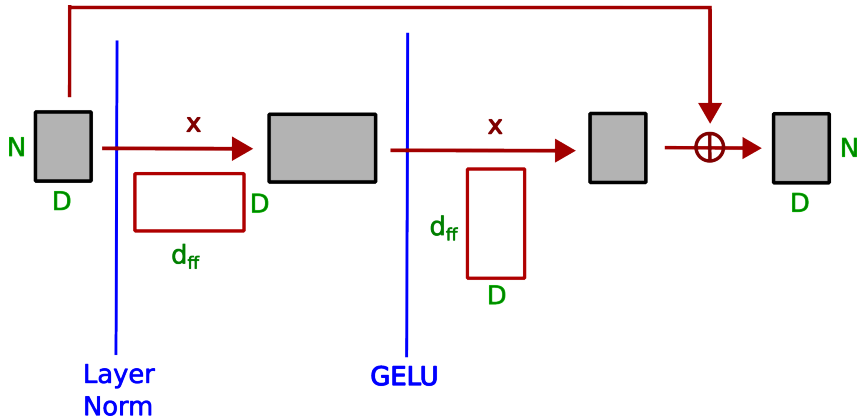
ViT: encoding



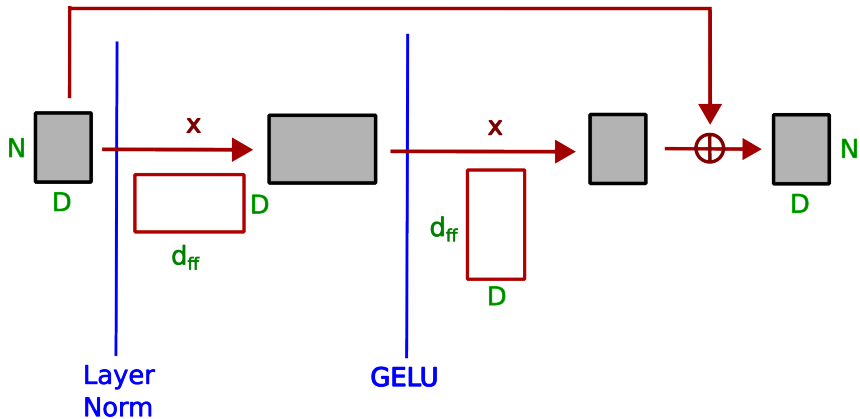
ViT: multi-head self-attention



ViT: multi-layer perceptron



ViT: multi-layer perceptron



The Multi-Head Self Attention and Multi-Layer Perceptron are iterated several times.

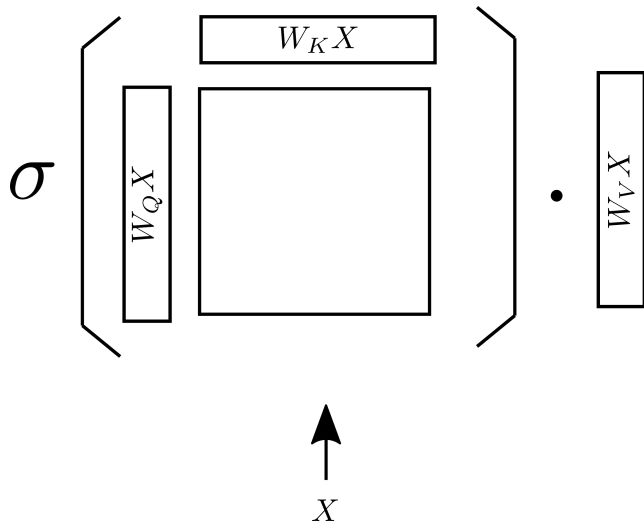
Scaled dot-product attention

Definition

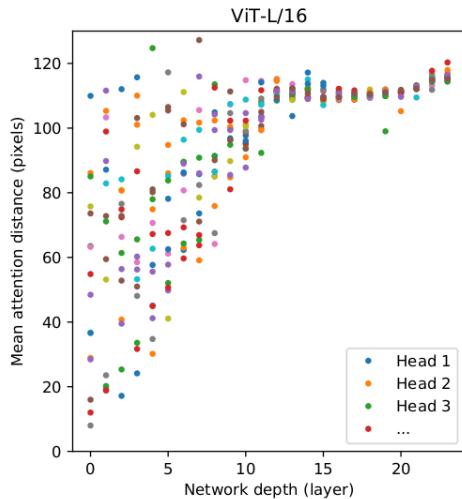
$$\text{Att}(Q, K, V) = \sigma \left(\frac{QK^t}{\sqrt{d_K}} \right) V$$

- V : values; K : keys; Q : queries.
- d_K is the length of K .
- σ : row-wise soft-max.

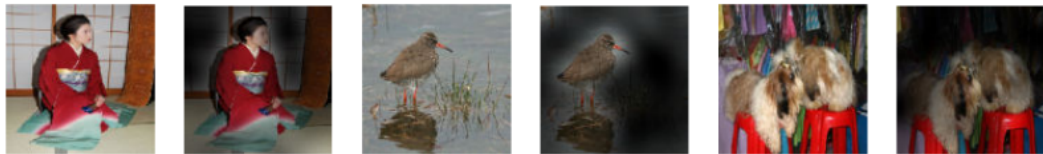
Self-attention



Mean attention distance

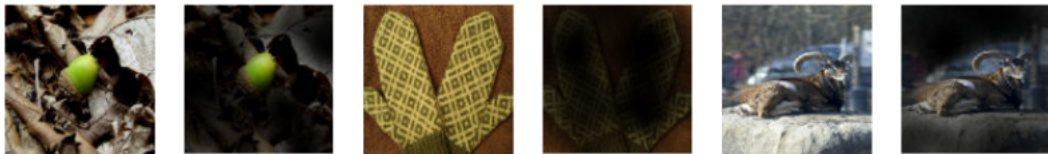


Interpretability



Method by Abnar et al. used to “integrate” attention [Abnar and Zuidema, 2020]

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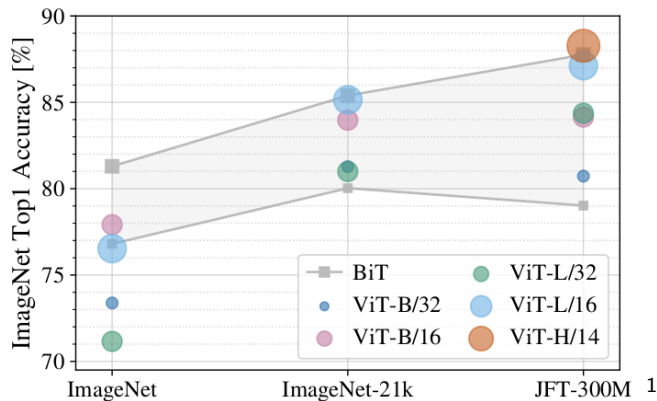
ViT results

	Ours-JFT (ViT-H/14)	Ours-JFT (ViT-L/16)	Ours-I21K (ViT-L/16)	BiT-L (ResNet152x4)	Noisy Student (EfficientNet-L2)
ImageNet	88.55 ± 0.04	87.76 ± 0.03	85.30 ± 0.02	87.54 ± 0.02	88.4/88.5*
ImageNet ReaL	90.72 ± 0.05	90.54 ± 0.03	88.62 ± 0.05	90.54	90.55
CIFAR-10	99.50 ± 0.06	99.42 ± 0.03	99.15 ± 0.03	99.37 ± 0.06	—
CIFAR-100	94.55 ± 0.04	93.90 ± 0.05	93.25 ± 0.05	93.51 ± 0.08	—
Oxford-IIIT Pets	97.56 ± 0.03	97.32 ± 0.11	94.67 ± 0.15	96.62 ± 0.23	—
Oxford Flowers-102	99.68 ± 0.02	99.74 ± 0.00	99.61 ± 0.02	99.63 ± 0.03	—
VTAB (19 tasks)	77.63 ± 0.23	76.28 ± 0.46	72.72 ± 0.21	76.29 ± 1.70	—
TPUv3-core-days	2.5k	0.68k	0.23k	9.9k	12.3k

Remarks

- Models pre-trained on JFT-300M
- Note the required processing power

ViT results



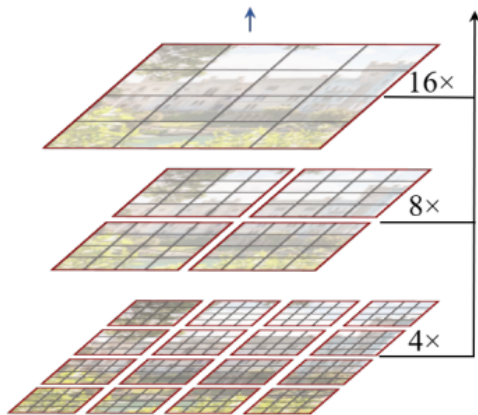
- ViT-H/14 requires 2500 TPUv3-core-days for pre-training
- But: “Training data-efficient image transformers & distillation through attention” [Touvron et al., 2021].

¹BiT: Big transfer [Kolesnikov et al., 2020]

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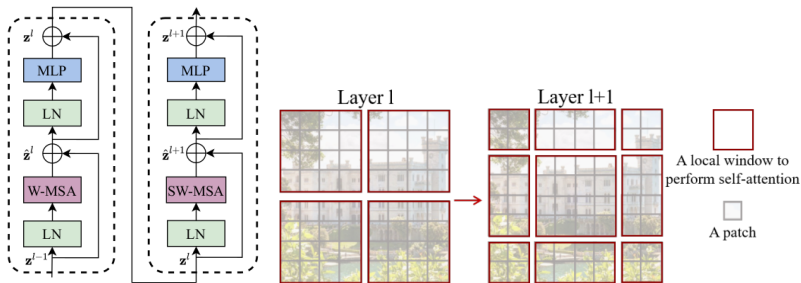
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Shifted window (SWIN) transformer [Liu et al., 2021]



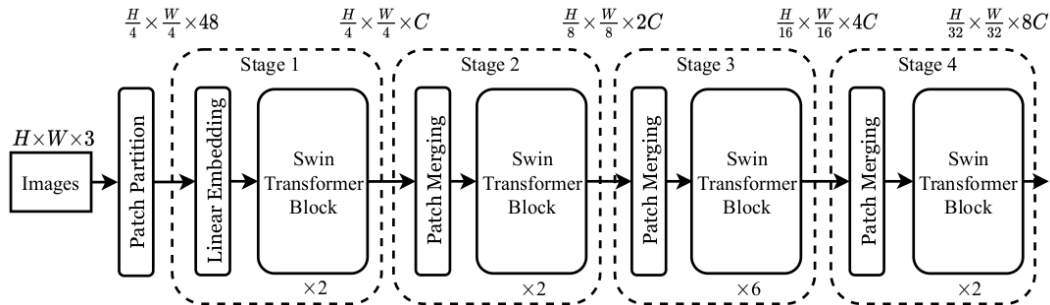
- The transformer modules are applied within each window
- Hierarchical approach: patches are merged at some levels
- The model uses **shifted windows**

SWIN blocks



- Multi-headed self-attention with regular (W-MSA) and shifted (SW-MSA) windowing configurations are applied alternatively

SWIN architecture



Results

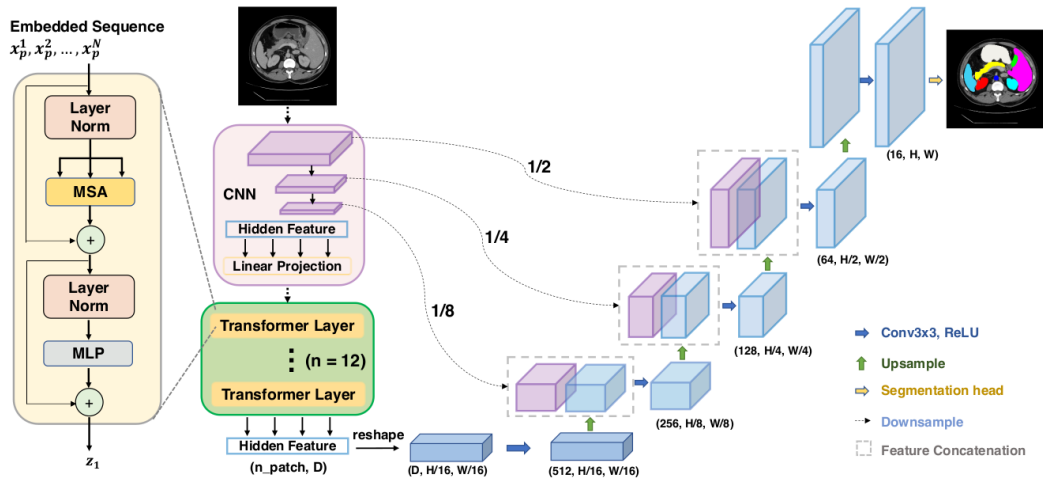
The Swin transformer obtains better results than previous methods on:

- ImageNet 1k and 22k classification
- COCO object detection and image segmentation
- ADE20k semantic segmentation

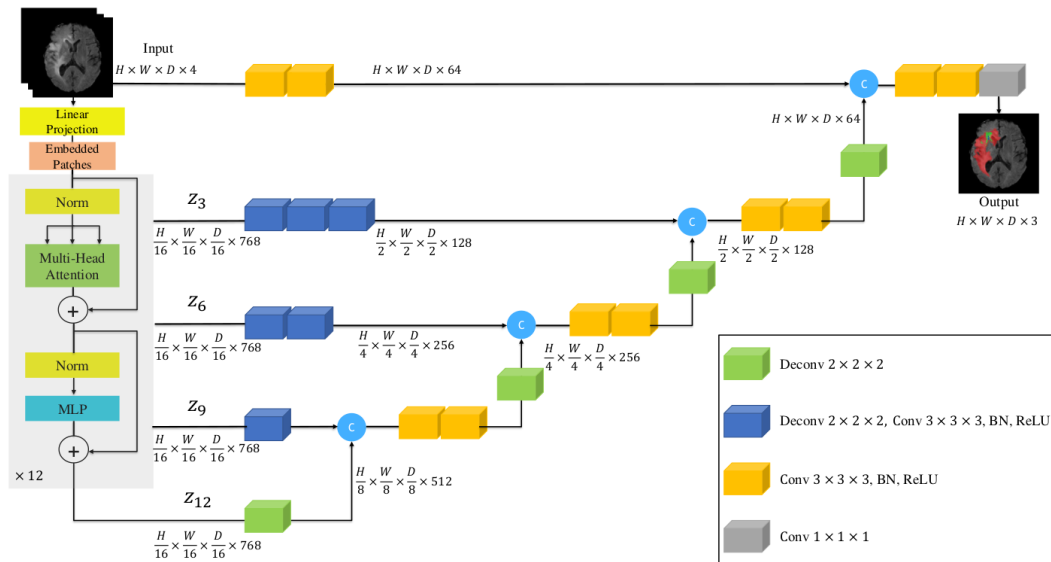
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TransUNet [Chen et al., 2021]



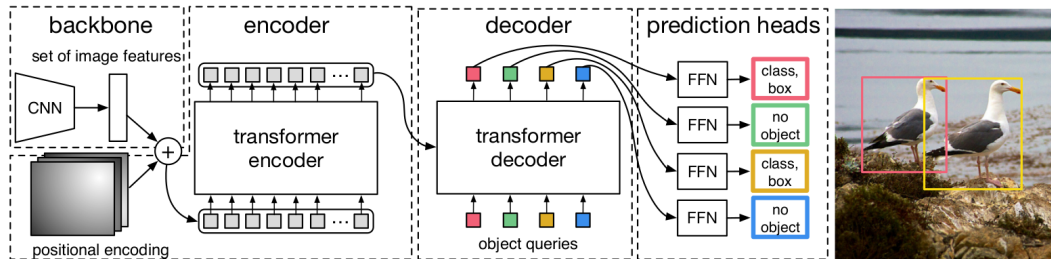
UNETR [Hatamizadeh et al., 2022]



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DETR: detection transformer [Carion et al., 2020]

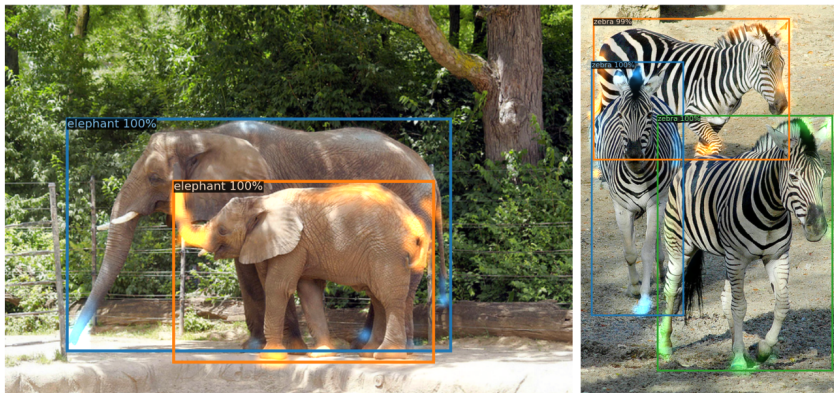


Remarks

- Convolutional layers are used to encode the image
- After a 1×1 convolutional layer, each feature map is flattened and considered as an input for the transformer encoder
- Decoder outputs are processed by a feed-forward network (FFN) to generate the box coordinates and label (possibly $\emptyset$).

Results and comments

- Similar accuracy and run-time performance to Faster R-CNN on the COCO object detection dataset
- Optimization was apparently difficult (extra losses, for instance)



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Task: segment the rabbits that are in danger



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Discussion

Convolutional neural networks

- Convolutional networks are based on two inductive biases:

Transformers

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Convolutional neural networks

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Transformers

- Transformers do not make any assumptions on the data structure

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Transformers

- Transformers do not make any assumptions on the data structure
 - Localization is brought by a positional encoding

Discussion

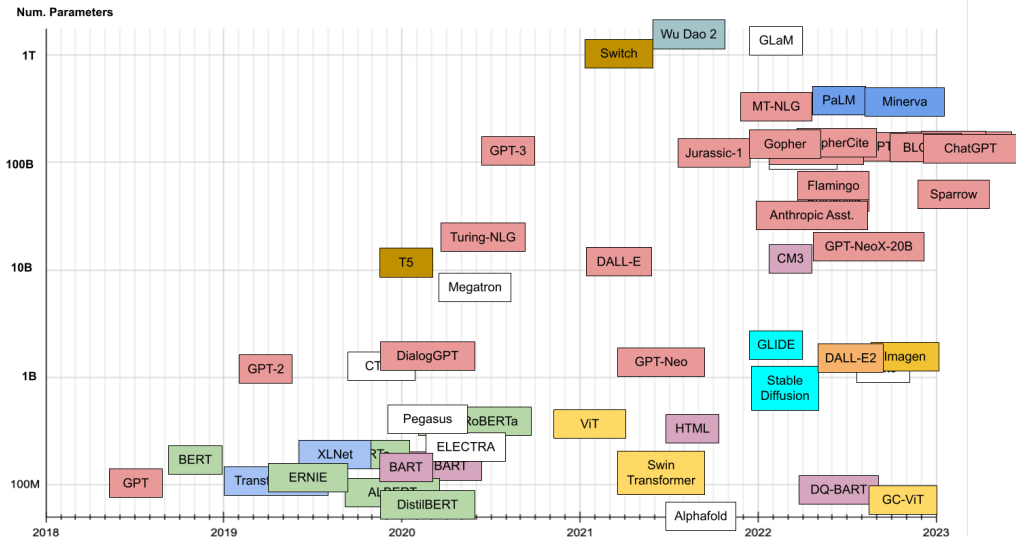
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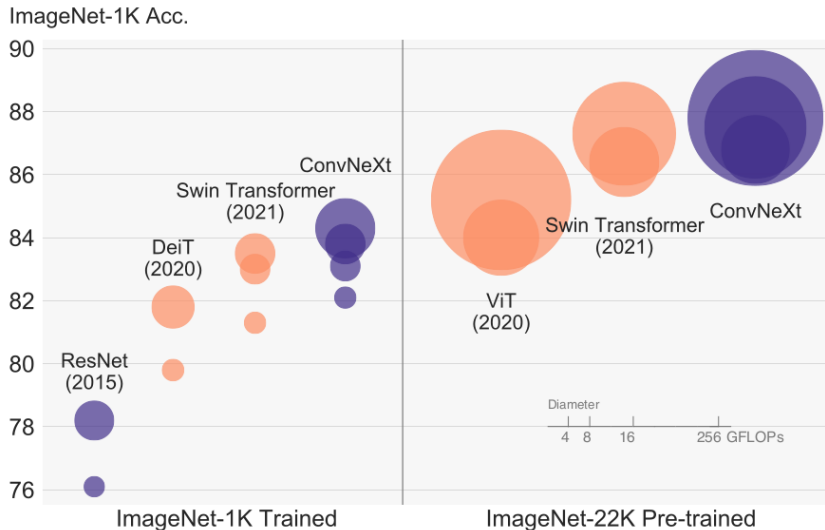
- Transformers do not make any assumptions on the data structure
 - Localization is brought by a positional encoding
- Are transformers a smart way of analysing images with fully connected layers?

Transformers chronology [Amatriain, 2023]



And the winner is ... [Liu et al., 2022]

And the winner is ... [Liu et al., 2022]



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